Learning Guide 4

2.3B Factors affecting the rate of dissolving

Name: _____

Class: _____ ( ) Date: _____

Objectives:

Knowledge: Recognize the factors affecting the rate of dissolving.

Skill: Perform fair tests to study the how volume of water affecting the rate of dissolving.
Use AI as a data analyst to evaluate data reliability and error sources.

Attitude: Take ownership of learning by predicting, verifying, and using AI coaching to deepen scientific reasoning.

Dictionary (with pronunciation)

★ Keywords

(I) **Underline** the **vowels** (a, e, i, o, u) and **break up the words** into **syllables** (only for the first word).

(II) Check the dictionary/textbook and write the **Chinese meaning** of the words in the brackets provided.

(III) Review the keywords and prepare for **dictation**.

Factors affecting the rate of dissolving			Diagrams
(a) size	(*noun*)	(大小)	/
(b) volume	(*noun*)	(體積)	/
(c) temperature	(*noun*)	(溫度)	/
(d) mass	(*noun*)	(質量)	/
(e) stir	(*verb*)	(攪拌)	
(f) increase	(*verb*)	(上升)	
(g) decrease	(*verb*)	(下降)	

Hints: 大小、溫度、體積、上升、下降、攪拌、質量

Preparation Questions

(Read textbook P.95–100 and answer the following questions)

(Check the Chinese meaning of the words highlighted before lesson)

Think-pair-share The Morning Drink

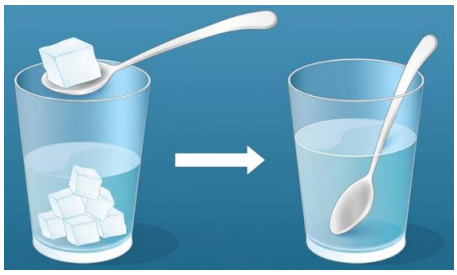
Q1. **Think**: You want to dissolve salt **faster**. What can you do?



I can dissolve salt faster by

_____

_____





Pair: Pair-up with your groupmates and share your thoughts.

Share: Discuss your methods with whole class.

Language Support:

If we say something **dissolves faster**, that means it has a **higher rate of dissolving**.

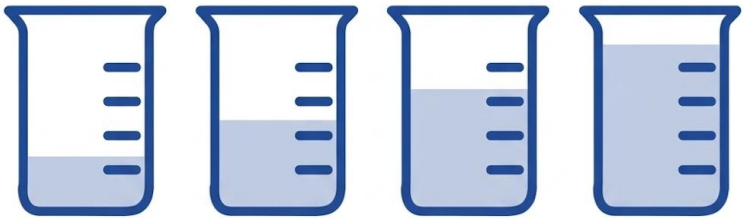
Pre-lab Session

Q2. We are going to carry out fair tests to study how **volume of water** affect the rate of dissolving.

Complete the table below.

Hypothesis	The rate of dissolving of salt in water increases (*increases / decreases*) if **more** water is used.
Variable to be changed	Volume of water
Variables to be compared	Time needed for the salt to completely dissolve in water
Variables to be kept constant (Put "✓" in the box(es))	☑ Temperature of water ☐ Volume of water ☑ Mass of salt ☑ Size of salt pieces

Experiment Time: **Experiment 2.5 – Factors affecting the rate of dissolving**

Step 1	Prepare 50 cm^3, 100 cm^3, 150 cm^3 and 200 cm^3 of water in four beakers of the same size.  50 cm^3 100 cm^3 150 cm^3 200 cm^3
Step 2	Add a spatula of salt and start the stopwatch. Stir the liquid with a glass rod until all salt dissolved. 
Step 3	Record the time taken for all the salt dissolved.

Results:

Volume of water	50 cm^3	100 cm^3	150 cm^3	200 cm^3
Time needed for the salt to completely dissolve in water	_____ s	_____ s	_____ s	_____ s

Input your data into **Agent Water**. He may ask you some interesting questions. Try to answer him and record what you have learnt through the conversation.

Key Learning Points Summary / 關鍵學習小結

Agent Water

1. _____
2. _____
3. _____

Conclusion:

Salt dissolves faster (*faster / slower*) if more volume of water is used.

Reflection:

High-order thinking questions:

Q1. Why do we **take average** of eight groups of the reading?

To reduce random errors – the results are more reliable.

Q2. Suggest **limitations** of this experiment and some methods to make the results **more reliable**.

1. Limitation: human error in manual stirring

   Improvement: let the same person stir the solution // use magnetic stirrer

2. Limitation: subjective judgment of dissolving endpoint

   Improvement: let the same person decide the endpoint // use digital light/turbidity sensors

3. Limitation: stopwatch reaction time lag

   Improvement: video recording the experiment for timestamping

Extension:

The rate of dissolving depends on many other factors. Try to design and carry out fair tests to study how different factors affect the rate of dissolving.

Factors	Comparison			Checklist	
Volume of water	more water	V.S.	less water	☑	(Done in this lesson)
Stirring	stirred	V.S.	not stirred	☐	
Size of the solute pieces	powdered salt	V.S.	salt cube	☐	
Temperature	hot water	V.S.	cold water	☑	(Done in Ch1)